DDA3003 Midterm Exercises Spring 26

Short answer & selections

InforVis Intro

1. What is *information visualization* interactive visual representation of abstract data to amplify cognition

2. "The purpose of visualization is insight, not pictures." What is "insight" ? List 2 items
   Find previously unknown relation or pattern; or help find what to ask when not having an idea.
3. What is EDA (Exploratory Data Analysis), and when is it most appropriate to use it ?
   an approach to analyzing datasets to summarize their main characteristics
   - Give one example of a question that might *emerge after* seeing a visualization.

E.g. draw scatter point of hunger-wasting data, asking 'Why the two scarcely appear in the same country at once'

4. *Name the two key challenges in InfoVis and briefly describe why each is challenging.*

Scale, Diversity: Data source too large to show, data too diverse to put into limited graph

5. Which statement best describes **InfoVis** in this course?
   A. Making attractive images from data
   B. A computer-supported interactive visual representation of abstract data to amplify cognition
   C. Turning spatial data into tables for analysis
   D. Using statistics to replace visual reasoning

6. Which is ***NOT*** listed as an "**insight**" of infovis?
   A. Discovery
   B. Decision making
   C. Explanation
   D. Encryption

7. Visualization is especially useful in:
   A. Confirmatory analysis only
   B. Exploratory Data Analysis (EDA)
   C. Data collection
   D. Hardware optimization

8. A visualization achieves its highest value when it:
   A. Uses 3D effects
   B. Forces us to see what we were not expecting
   C. Includes as many icons as possible
   D. Avoids interaction

9. Which of the following is an example of "**chart junk**" mentioned in the lecture?
   A. Baselines
   . 3D
   C. Labels
   D. Axes

10. Which is ***most** aligned* with "**graphical integrity**"?
    A. Manipulating axes to make differences look bigger
    B. Choosing encodings that "tell the truth" and avoid distortion
    C. Adding decorative shapes to attract attention
    D. Hiding the baseline to save space

11. T/F: InfoVis is mainly about producing a graphic or image; cognition is secondary.

12. T/F: EDA is most useful when you don't yet know what you're looking for.

13. T/F: Interaction is a minor detail; representation is the only key aspect.

14. T/F: "Maximize data-ink ratio" implies removing unnecessary decoration.

15. T/F: Sometimes a table is better than a graph.

Graph design principles

1. List four characteristics of "Excellent Graphics" from the lecture.
2. Which is *most directly* about avoiding misleading magnitude comparisons in bar charts?
   A. Use rainbow colors
   B. Always remove axes
   C. Use a zero baseline (when length encodes magnitude)
   D. Add 3D to show depth
3. Which best matches "multifunctioning graphical elements"?
   A. Decorative icons to increase memorability
   B. One graphical mark encodes multiple pieces of information
   C. Using separate legends for every variable
   D. Using 3D to encode extra variables

4. Which design best supports *macro/micro readings*?
   A. A single number KPI with no context
   B. A chart that shows overview and allows seeing fine details on the same page
   C. A heavily annotated infographic with no data scale
   D. A 3D pie chart with exploded slices

5. Which are part of the “Excellent Graphics” list?
   - ☐ Present many numbers in a small space
   - ☐ Encourage the eye to compare different pieces of data
   - ☐ Reveal data at several levels of detail
   - ☐ Prioritize decoration over content
   - ☐ Integrate with statistical and verbal descriptions

6. Which are listed under Tufte’s principles?
   - ☐ Avoid chart-junk
   - ☐ Small multiples
   - ☐ Macro/Micro readings
   - ☐ Appropriate applications of color
   - ☐ Randomize color to increase “appeal”

7. Which are recommended actions to maximize data-ink ratio?
   - ☐ Erase non-data-ink
   - ☐ Erase redundant data-ink
   - ☐ Revise and edit
   - ☐ Add drop shadows for depth
   - ☐ Add background textures to “tell a story”

8. T/F: Scatter Plot can reduce certain bar-chart distortions because they don’t rely on comparing bar lengths.

D3

11. `selectAll()` differs from `select()` because `selectAll()` :
    A. selects only the first matching element
    B. selects all matching elements
    C. can’t use CSS selectors
    D. only works for SVG elements

12. Why do you “only see one circle” if all joined circles share the same `cx`, `cy`, and `r`?
    A. D3 only creates one SVG element per chart
    B. SVG cannot draw more than one circle
    C. All circles overlap at the same location with the same size
    D. `join()` automatically removes duplicates

13.

entity, information of that entity, and relation between attributes or entities.
E.g. Carlos is an object, age is an attribute, 23 yrs and Carlo has relationship of name and age.

Multivariate data & Charts

1. what are *objects*, *attributes*, and *relations*? Give one example of each for a "students/courses" dataset.
2. What is a *case* in a data table? What is a *variable* (attribute/dimension)?
   each horizontal row is a case, variable is in each column.
3. Explain the difference between a *wide table* and a *long table* (use the CO_2 example idea). wide table: each attribute gets a column, long table: each data case has an attribute-value pair
4. List the three variable types: *nominal*, *ordinal*, *quantitative*, and give one example of each. E.g.: name, clothes_size, GPA
5. What is *metadata* and why is it useful when "the table itself isn't enough"?
   Descriptive data to help understand data.
6. List two common data issues that require cleaning.
   Typos, data conflict, syntax error, white space, etc.
7. You should prefer a *table* over a graph when you need to:
   A. Reveal relationships among values
   B. See the overall "shape" or trend
   C. Look up precise individual values
   D. Emphasize patterns over exact numbers

8. You should prefer a *graph* over a table when:
   A. The message is contained in the *shape* of the values
   B. You must report exact values for each row
   C. The dataset has only one row
   D. You want to maximize the number of decimals shown

9. Which is an example of an *ordinal* variable?
   A. Country code (CHN, USA)
   B. Income level (low, middle, high)
   C. Temperature (°C)
   D. Population size

10. Which best describes the main purpose of *data cleaning*?
    A. Make the dataset larger
    B. Ensure data is correct, consistent, and usable for analysis/visualization
    C. Replace graphs with tables
    D. Add more metadata fields automatically

11. Which items are listed as potential *data issues*?
    □ Missing fields
    □ Duplicates
    □ Conflicting records
    □ Delimiters/white spaces
    □ "Too many colors in a chart"

12. Which are listed as steps in the "How to clean your data" process?
    □ Remove unwanted/irrelevant observations

□ Fix structural errors
□ Standardize your data
□ Remove unwanted outliers (only if erroneous)
□ Hide contradictory errors

13. T/F: A 3D scatter plot is always the best choice for trivariate data.
14. T/F: For trivariate data, it's often better to stay in 2D and encode the third variable using a mark property (e.g., color/shape).
15. T/F: Scatterplot matrices show every pairwise relationship, but can still miss higher-order (multi-variable) patterns.
16. For a *line chart*, what kinds of patterns in time series data is it typically used to show? (List at least 2.) Trend, Exception, Change rate, Cycles.
17. Why is it especially important that *bar charts start at a zero baseline*?
    For integrity: since we read the length of bar as data,
    other baseline will cause distortion and the proportion of data and graph to be non-linear.
18. You have long category labels and many items to compare. Which is usually better?
    A. Vertical bars
    B. Horizontal bars
    C. 3D bars
    D. Radar chart

19. When values span ***different orders of magnitude*** and you want to compare growth rates (e.g., both 10%), which scale may be better?
    A. Log scale
    B. Random scale
    C. Start y-axis at 0 always
    D. Use a pie chart

20. Which chart is primarily for showing *individual contributors plus an increasing total*?
    A. Strip plot
    B. Pareto chart
    C. Box plot
    D. Bump chart

21. Which are ***distribution* views** listed in the lecture?
    □ Box plot □ Histogram □ Dot plot □ Strip plot □ Stem-and-leaf plot □ Trellis display

22. Which are listed strategies to address ***overplotting***?
    □ Jitter points
    □ Make points transparent
    □ Aggregate or sample data
    □ Encode density
    □ Add more 3D effects

23. Which time-series patterns are listed as important to show?
    □ Trend □ Variability □ Rate of change □ Cycles □ Exceptions

24. T/F: Unlike bar charts, a line chart's y-axis does not always need to start at 0, and this can be acceptable if it doesn't mislead the viewer about magnitude.

25. T/F: If you have many lines in one line chart, you can use *small multiples* to improve readability.

Graphs / Networks

1. What are two main ways to represent a graph?
   Adjacency matrix and list.
2. Given two real-world systems that can be modeled as graphs.
   Internet, railway system, transaction record, LinkedIn relationship, etc.
3. List the benefits of *filtering* in graph visualization and explain what filtering does *not* do to the underlying data.
   Focus on relevant parts, Reveal patterns or communities, support analysis, improve visibility.
   It does NOT delete the data
4. List the interactions you may have with a graph visualization.
   Zoom and pan, annotation, identify, paths, filter, neighbor, isolate, export, legend, etc.
5. Which visualization type is most appropriate for *flows* (as listed)?
   A. Scatterplot matrix
   B. Sankey diagram / chord diagram
   C. Box plot
   D. Histogram

6. Which statement best matches the warning about layout perception ?
   A. Layout never affects perception of centrality
   B. A node placed in the center may be perceived as more central even if statistics don't support it
   C. Centrality only depends on node color
   D. Triangles cannot appear in graphs

7. Which is a key use of *time-oriented* layouts?
   A. Identify node labels more clearly by color
   B. Facilitate seeing sequence/earlier vs later nodes when time is part of the graph
   C. Compute eigenvector centrality
   D. Ensure graphs have a single component

8. Which interaction is most directly used to "inspect detailed data" about a node?
   A. Drag
   B. Identify (hover/click)
   C. Export
   D. Explanation sequence

9. Which is the best first step when a graph is “too dense to see any edges” after filtering?
   A. Add more edges back
   B. Redo the layout (e.g., force-directed/Force Atlas 2)
   C. Remove node attributes
   D. Convert to an adjacency matrix

10. Which task is most aligned with *Sankey* diagrams?
    A. Ranking changes over time
    B. Showing volume of flow through a multi-step process
    C. Showing a distribution of values
    D. Showing pairwise correlations

11.

12. Which are listed as “basic graph statistics” in the lecture?
    □ Density □ Components □ Degree □ Path □ Centrality □ Hue

13. Which layouts are listed in the lecture?
    □ Force-directed □ Node-only □ Time-oriented □ Top-down □ Radial □ Treemap □ Hierarchical pie □ Sunburst □ Maps □ Chord □ Parallel coordinates

14. Which are listed as *essential visual attributes*?
    □ Node size □ Node color □ Label □ Edge weight □ Edge color □ Edge type

15. Which interactions are explicitly suggested in the “Interact” step?
    □ Zoom
    □ Select
    □ Filter
    □ Annotate / explain
    □ Randomize node positions

16. Which approaches to visualize/analyze graphs are listed?
    □ node-link relationships
    □ Hierarchies (trees, sunburst)
    □ Communities (clustered layouts)
    □ Flows (Sankey/chord)
    □ Only bar charts and line charts

17. Which interactions are listed in the lecture?
    □ Zoom & pan
    □ Identify (hover/click)
    □ Filter
    □ Isolate
    □ Neighbors
    □ Paths
    □ Annotate
    □ Export

18. T/F: Filtering is equivalent to deleting data, so you cannot restore the hidden edges/nodes.
19. T/F: Ego-network analysis focuses on one node, its neighbors, and the connections among those neighbors.
20. T/F: When filtering/drilling down or changing layout, you must clearly convey the change of context to avoid confusing viewers.
21. T/F: The graph visualization process includes data preparation, layout, adding visual attributes, and interaction/analysis.
22. T/F: A top-down hierarchy layout can become too wide, making node labels difficult to place.
23. T/F: When data includes positive and negative values, it is better to use size for the sign and color for the magnitude.
24.

Visualization Designs

1. *Choose a representation + sketch (matrix vs list).*
   Given a small directed graph with vertices {A,B,C,D} and edges {A→B, A→C, C→A, C→D}.
   - Sketch the *adjacency matrix* (with row/column labels) and the *adjacency list*.
   - Circle one edge and show how each representation indicates its presence.
2. *Business network dashboard (node-link + attributes).*
   Design a sketch for a *supply chain* network view: facilities as nodes, shipping routes as edges.
   - Encode *inventory/capacity/utilization* on nodes using at least two visual variables (size, color, icon).
   - Encode *average cost* on edges using thickness or color.
   - Include a small legend or direct labels.
3. *Flow visualization selection (Sankey vs chord).*
   Sketch a scenario with 4 entities (e.g., departments or states) and flows between them.
   - Create a rough *Sankey* sketch and a rough *chord* sketch for the same data.
   - Under each sketch, write one sentence describing what question it helps answer best.
4. *Time-oriented vs force-directed (two sketches).*
   Given a patent-citation graph where time matters, sketch:
- (A) a force-directed layout that hides time ordering,

- (B) a time-oriented (hierarchical) layout where earlier nodes are above later nodes. Write 1 sentence on what insight (B) reveals that (A) may obscure.

24. *Choose an appropriate layout (design decision).*
    For each scenario, draw a rough layout choice and label why:

- (i) airline routes with airport latitude/longitude
- (ii) trade flow between countries (directional)
- (iii) bipartite graph (students ↔ courses) where you want to compare multiple attributes across nodes
  (Use maps / chord / parallel coordinates as appropriate.)

21. *Interaction menu design (sketch UI).*
    Sketch a graph visualization interface (like a dashboard) that supports at least *6* of the interactions from the lecture.
    Your sketch must include:

- where zoom/pan happens,
- how “identify” is triggered (hover tooltip or click side panel),
- a filter panel with at least two filters (giant component, edge weight),
- an annotation mechanism,
- an export button.

5.